



Reversible Computing

Michael P. Frank, as a graduate at MIT, designed one of the first universal DNA computers, and discovered that it had to be reversible for fundamental thermochemical reasons. Since his graduation from MIT in 1999, Mike has been continuing his reversible computing research as an Assistant professor at the University of Florida's Department of Computer and Information Science and Engineering. *Digit* caught up with him to speak about reversible computing.

■ What is reversible computing?

Reversible computing—revcomp for short—is the application of principles of recycling to computing. When a computer performs a logical operation, the unwanted bits after the operation are thrown away, and dissipated as heat. There is a limit on how far we can keep doing this. If we want computer speeds to keep on increasing, we need to 'uncompute' the unwanted bits.

Reversible computing means computing using a physical mechanism that is thermodynamically reversible—one that ballistically coasts along through its computation with low friction. Thus, it will dissipate only a small fraction of a bit's energy to heat with each logic operation.

If a computation is thermodynamically reversible, it would need to be logically reversible as well. Performing any desired computation is still possible, despite this constraint. But it strongly affects digital logic designs. In the long term, it will impact instruction sets and high-level programming languages as well. Eventually, these will also have to be reversible to provide optimal efficiency.

■ Why do we need reversible computing?

With device sizes fast approaching atomic-scale limits, more energy efficiency is essential. Energy efficiency fundamentally affects the speed of circuits—such as nanocircuits—and therefore the speed of most computing applications. Ballistic circuits that conserve information, by uncomputing bits instead of throwing them away, will soon offer the only physically possible way to keep improving energy efficiency.

■ What is a ballistic circuit?

A ballistic process is a process in which a system proceeds forward under its own momentum, with only a small fraction of the free energy in the system being lost to heat. For example, when an ICBM (Inter-Continental Ballistic Missile) makes a flight between continents, it 'coasts' (movement of a vehicle with hardly any generation of heat) for most

of its journey, with only a small portion of its potential and kinetic energy being lost to friction during its trajectory through the atmosphere.

■ How soon do you think revcomp is going to become important?

It will become important when the chip-making industry runs out of other tricks for reducing the energy cost of logic operations. Throughout the history of the industry, it has been reducing the energy cost of logic operations by reducing bit energies; but, there are some absolutely irrevocable limits to this process. These limits will be reached within the next three decades. This has to do with the fact that the density of all of the physical information that is present in materials at normal temperatures and pressures is finite, and moreover, it is quite limited—at most, a small handful of bits per atom. We can never store a logical bit using less than 1 physical bit, and the irreversible transformation of a physical bit results in heat energy that must be expelled into the environment. In room-temperature environments, this results in a strict maximum of about 300 billion billion irreversible bit-operations that can be performed per Joule of energy consumed. This may sound like a lot, but we are only about a factor of 1,000,000 below this point. If past trends continue, we would expect to reach this point in only another 25 years, or so.

■ Why do we need to bother about it now?

To make reversible computing practical, there are many difficult engineering research problems that remain to be solved, such as the design of extremely high-quality oscillators. If we don't start working hard on developing solutions for these now, there is a significant danger that solutions will not be ready when traditional approaches run out of steam. If this happens, the development of computer technology will stall—no longer will we be able to expect that every year we will be able to get a more powerful processor with lower power consumption, which can be utilised for new applications not previously possible.

This can have a significant braking effect on the growth of the entire world economy, since productivity growth in many industries today relies largely on the increasingly extensive application of information technology. The rate at which IT can penetrate into new application areas will drastically slow down, if the performance-per-unit-power-consumption of digital technology stops improving.

// Our civilisation will depend on exploring the problems that reversible computing poses //

